

# Digital Evidence & Forensic Incident Dossier

|                      |                                                                        |                  |                                            |
|----------------------|------------------------------------------------------------------------|------------------|--------------------------------------------|
| REPORT REFERENCE:    | RPT-█20260908-█2944A6                                                  | GENERATED (UTC): | 2026-█09-█08 10:█16:█42 UTC                |
| EVIDENCE IDENTIFIER: | EVD-█5E58C282C0                                                        | ASSIGNED CASE:   | UNASSIGNED /█ DIRECT EVIDENCE              |
| INVESTIGATOR:        | Lead Forensic Examiner                                                 | CLASSIFICATION:  | CONFIDENTIAL // EVIDENTIARY DISCLOSURE     |
| EMAIL SHA-256:       | 5e58c282c0ac76456bf139f43214█0e0d<br>fc5338ca13502c9bea990d35█aa825c3b | HASH ALGORITHM:  | FIPS PUB 180-4 SHA-256 (Preserved Literal) |

## 1. Executive Forensic Summary

|                            |                   |                         |                   |
|----------------------------|-------------------|-------------------------|-------------------|
| DETERMINISTIC THREAT SCORE | ASSESSED SEVERITY | ML PHISHING PROBABILITY | ML CLASSIFICATION |
| 38/100                     | SUSPICIOUS        | 65.5%                   | PHISHING          |

Email risk assessed as Suspicious (38/100) driven by 3 forensic signal(s): Possible PayPal lookalike domain (+18), Suspicious URL detected (+8), Credential harvesting or urgent security language (+12).

## 2. Email Headers & Sender Alignment

|              |                                                   |
|--------------|---------------------------------------------------|
| From:        | security-█alert@█paypal-█update.█com              |
| To:          | victim@█company.█com                              |
| Subject:     | Urgent Security Alert:█ Account Suspicious Access |
| Date:        | Tue,█ 08 Sep 2026 10:█30:█00 +█0000               |
| Reply-To:    | None                                              |
| Return-Path: | None                                              |
| Message-ID:  | &█lt;█evd-█test-█12345@█paypal-█update.█com&█gt;█ |

|                     |                     |                    |                        |
|---------------------|---------------------|--------------------|------------------------|
| From Domain:        | paypal-█update.█com | Reply-To Domain:   | N/█A (Aligned)         |
| Return-Path Domain: | N/█A (Aligned)      | Alignment Verdict: | Sender domains aligned |

## 3. Email Authentication Findings (SPF / DKIM / DMARC)

|                       |                  |                                                                |
|-----------------------|------------------|----------------------------------------------------------------|
| PROTOCOL              | OBSERVED VERDICT | TECHNICAL DETAILS & EVIDENCE                                   |
| SPF (Sender Policy)   | NONE             | No Authentication-█Results or Received-█SPF header present.█   |
| DKIM (Cryptographic)  | NONE             | No Authentication-█Results or DKIM-█Signature header present.█ |
| DMARC (Domain Policy) | NONE             | No Authentication-█Results header present.█                    |

Notice: Observed authentication result from supplied headers (unverified by local mail server)

## 4. Deterministic Threat Score Breakdown

|          |         |                                  |
|----------|---------|----------------------------------|
| CATEGORY | IMPACT  | FORENSIC FINDING & JUSTIFICATION |
| General  | +18 pts | N/A                              |
| General  | +8 pts  | N/A                              |

|         |         |     |
|---------|---------|-----|
| General | +12 pts | N/A |
|---------|---------|-----|

5. NLP Content Phishing Classification

|                       |                                                 |                   |                                      |
|-----------------------|-------------------------------------------------|-------------------|--------------------------------------|
| Model Name:           | TF-IDF + Logistic Regression                    | Classification:   | PHISHING                             |
| Phishing Probability: | 65.5%                                           | Model Confidence: | medium                               |
| Top Text Features:    | httpurl, account, immediately, security, urgent | Signal Role:      | Auxiliary ML signal (Non-overriding) |

6. AI Analyst Assistant Synthesis

|                     |                                |                   |                                                                                   |
|---------------------|--------------------------------|-------------------|-----------------------------------------------------------------------------------|
| Likely Attack Type: | Credential Harvesting Phishing | Likely Objective: | Theft of corporate directory or single-sign-on credentials via spoofed interfaces |
|---------------------|--------------------------------|-------------------|-----------------------------------------------------------------------------------|

**Executive Synthesis:** The analyzed email presents characteristics of Credential Harvesting Phishing with a deterministic threat score of 38/100 (SUSPICIOUS). Key driving indicators include 3 forensic flag(s). Adversary objective is assessed as: Theft of corporate directory or single-sign-on credentials via spoofed interfaces.

- Key Forensic Evidence Identified by LLM:**
- Global threat score evaluated at 38/100 (SUSPICIOUS severity).
  - Forensic indicator: Possible PayPal lookalike domain
  - Forensic indicator: Suspicious URL detected
  - Forensic indicator: Credential harvesting or urgent security language
  - Brand lookalike detected: 'paypal-update.com' targeting 'PayPal'.

7. Extracted Indicators of Compromise (IOCs)

Observed URLs & Suspicion Scoring:

| EXTRACTED URL / TARGET HOST    | SCORE  | FORENSIC OBSERVATIONS & FLAGS                                                                                                      |
|--------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------|
| http://paypal-update.com/login | 30/100 | Potential brand impersonation: similar to paypal.com (90% match), Suspicious security/authentication keywords found: login, update |

Observed Attachments & Cryptographic Checksums:

No file attachments detected in this message.

8. Network Transmission Path & Topology

|                                 |                 |                     |         |
|---------------------------------|-----------------|---------------------|---------|
| Earliest Observable Sending IP: | None identified | Reverse DNS / Host: | Unknown |
|---------------------------------|-----------------|---------------------|---------|

No sequential relay hops reconstructed from Received: headers.  
*Notice: Labeled strictly as observed infrastructure route. Does not represent physical attacker travel path.*

9. Infrastructure & Domain Intelligence

No public IP intelligence enriched for this email.

10. Brand Impersonation & Lookalike Domains

| SUSPECTED BRAND | SPOOFED DOMAIN      | SIMILARITY | TECHNIQUES & ATTACK PROFILE                 |
|-----------------|---------------------|------------|---------------------------------------------|
| PayPal          | paypal-█update.█com | 90.0%      | hyphenation,█ brand_█keyword,█ added_█affix |

11. Prescriptive Recommendations & Containment Actions

- [ ] 1. Tag message header with [SUSPICIOUS] warning banner for internal recipient.
- [ ] 2. Monitor user account for anomalous authentication or link traversal events.

12. Evidentiary Limitations & Chain-of-Custody Certification

**Forensic Chain-of-Custody Certification:**  
This document is generated by MailTrace AI based on literal RFC-822 evidence headers. Public IP geolocation represents coarse regional estimates based on RIR allocations and does not establish street-level physical locations or the personal identity of the sender. All cryptographic hash sums (SHA-256) were computed over unaltered message byte streams and preserve complete digital non-repudiation chain of custody.

|                           |                                                                                   |                      |                               |
|---------------------------|-----------------------------------------------------------------------------------|----------------------|-------------------------------|
| EVIDENTIARY DIGITAL SEAL: | SHA256:5e58c282c0ac76456bf139f43214<br>█0e0dfc5338ca13502c9bea990d35█aa825<br>c3b | VERIFICATION STATUS: | CRYPTOGRAPHICALLY<br>VERIFIED |
|---------------------------|-----------------------------------------------------------------------------------|----------------------|-------------------------------|